

# BML: MCMC

<http://github.com/rbardenet/bml-course>

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- 2** Monte Carlo methods
- 3** The Metropolis-Hastings algorithm
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What comes to *your* mind when you hear "Monte Carlo"?

### Minimizing the posterior expected loss

If  $\mathcal{A} = \{a_g\}$  and we partition  $s = (s_{\text{obs}}, s_u)$ , then, given  $s_{\text{obs}}$ , we choose

$$g^*(s_{\text{obs}}) = \arg \min_{g \in \mathcal{A}} \mathbb{E}_{s_u | s_{\text{obs}}} L(a, s).$$

### The bottleneck is computing integrals w.r.t. the posterior

- ▶ E.g. for binary prediction with 0-1 loss

$$y^* \in \arg \max_{y \in \{0,1\}} \int p(y|x, \theta) p(\theta | x_{1:n}, y_{1:n}) d\theta$$

- ▶ or for estimation with squared loss

$$\theta^* = \int \theta p(\theta | y_{1:n}) d\theta.$$

Let  $\pi$  be a pdf w.r.t.  $d\theta$ .

## The problem of numerical integration

Find  $T$  nodes  $(\theta_t)$  and weights  $(w_t)$  so that

$$\int f(\theta)\pi(\theta)d\theta \approx \sum_{t=1}^N w_t f(\theta_t), \quad \forall f \in \mathcal{C},$$

where  $\mathcal{C}$  is a large class of functions.

## A constraint for Bayesians: $\pi$ is only known up to a constant

E.g. in estimation,

$$\pi(\theta) = p(\theta|y_{1:n}) \propto p(y_{1:n}|\theta)p(\theta) =: \pi_u(\theta).$$

Or in classification/regression,

$$\pi(\theta) = p(\theta|x_{1:n}, y_{1:n}) \propto p(y_{1:n}|x_{1:n}, \theta)p(\theta) =: \pi_u(\theta).$$

- ▶ For modern developments, see quasi-Monte Carlo integration (Dick and Pillichshammer, 2010).

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## The Monte Carlo principle

Find a distribution on  $\theta_1, \dots, \theta_T$  and weights  $w_t$  such that

$$\mathcal{E}_T(f) = \sum_{t=1}^T w_t f(\theta_t) - \int f(\theta) \pi(\theta) d\theta$$

is small (with large probability, in quadratic mean, converges in law at some rate, etc.)

- If you knew how to sample from  $\pi$ , you could take  $\theta_t \sim \pi$  i.i.d.,  $w_t = 1/T$ , and prove e.g.

$$\mathbb{P} \left( \mathcal{E}_T(f) \geq \alpha \frac{\sigma(f)}{\sqrt{T}} \right) \leq \frac{1}{\alpha^2}, \quad \forall \alpha,$$

as soon as  $\sigma(f)^2 := \mathbb{V}_\pi[f(\theta) - \int f(\theta) \pi(\theta) d\theta] < +\infty$ .

- ▶ Let  $\pi_u(\theta) = Z\pi(\theta)$  be the unnormalized target pdf.
- ▶ Sample  $\theta_{1:T}$  i.i.d. from  $q$ , and take

$$w_t = \frac{\pi_u(\theta_t)}{q(\theta_t)} \times \left( \sum_{t=1}^T \frac{\pi_u(\theta_t)}{q(\theta_t)} \right)^{-1}$$

so that  $\sum w_t = 1$ .

- ▶ Then
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- ▶ One can show that  $\sqrt{T}\mathcal{E}_T(f) \rightarrow \mathcal{N}(0, \sigma_{\text{NIS}}^2(f))$ .
  - ▶ Problem is that for reasonable choices of  $f, q, \pi$ ,  $\log \sigma_{\text{NIS}}(f) \propto d$ .

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## (Mostly) friendly faces



**Figure:** A few MCMC pioneers: N. Metropolis, S. Ulam, A. Rosenbluth, W. K. Hastings

- ▶ The idea is to take  $(\theta_t)$  to be an ergodic Markov chain with limiting distribution  $\pi$ , so that for  $f \in L^1(\pi)$ ,
- ▶ In MCMC research, when a new Markov kernel comes out, we typically first prove a **law of large numbers**, and then a **central limit theorem**, i.e., that under weak conditions on  $\pi$  and  $f$ ,

and that  $\sigma^2(f)$  can be estimated; see (Douc, Moulines, and Stoffer, 2014).

Let  $(\theta_t)_{t \in \mathbb{N}}$  be a Markov chain on  $\Theta$ , with Markov kernel  $P$ . If

- There exists  $\pi$  s.t.

$$\int d\pi(\theta) P(\theta, B) = \pi(B).$$

- For any  $A$  with  $\pi(A) > 0$ , for any  $\theta \in \Theta$ ,

$$\mathbb{P}_\theta \left( \sum_{t=0}^{\infty} 1_{\theta_t \in A} = +\infty \right) = 1,$$

then for any  $f$  such that  $\int |f| d\pi < \infty$ , for any initial distribution  $\mu_0$  of  $\theta_0$ , almost surely

$$\frac{1}{T} \sum_{t=1}^T f(\theta_t) \rightarrow \int f d\pi.$$

See e.g. (Douc, Moulines, and Stoffer, 2014).

# The Metropolis-Hastings algorithm

MH( $\pi_u$ ,  $q(\cdot|\cdot)$ ,  $\theta_0$ ,  $T$ )

1       **for**  $t \leftarrow 1$  **to**  $T$

2            $\theta \leftarrow \theta_{t-1}$

3            $\theta' \sim q(\cdot|\theta)$ ,  $u \sim \mathcal{U}_{(0,1)}$ ,

4            $\alpha = 1 \wedge \frac{\pi(\theta')}{\pi(\theta)} \frac{q(\theta|\theta')}{q(\theta'|\theta)}$ .

5           **if**  $u < \alpha$ ,

6                $\theta_t \leftarrow \theta'$         $\triangleright$  *Accept*

7           **else**  $\theta_t \leftarrow \theta$         $\triangleright$  *Reject*

8       **return**  $(\theta_t)_{t=1,\dots,N_{\text{iter}}}$

... is given by

$$P_{\text{MH}}(\theta, \theta') = \alpha(\theta, \theta') q(\theta' | \theta) + \delta_{\theta}(\theta') \left[ 1 - \int \alpha(\theta, \vartheta) q(\vartheta | \theta) \right] d\vartheta,$$

where

$$\alpha(\theta, \theta') = 1 \wedge \frac{\pi(\theta')}{\pi(\theta)} \frac{q(\theta | \theta')}{q(\theta' | \theta)}.$$



- ▶ We first show detailed balance, i.e.,  $\pi(\theta)P(\theta, \theta') = \pi(\theta')P(\theta', \theta)$ .
- ▶ We deduce that  $P$  leaves  $\pi$  invariant.

### Theorem (Robert and Casella, 2004)

If  $\pi(A) > 0 \Rightarrow (\forall x)q(A|x) > 0$ , then  $P_{\text{MH}}$  satisfies the LLN.

### Some additional useful properties

- ▶ Note that if  $P_1$  and  $P_2$  leave  $\pi$  invariant, then so does

$$P_1 P_2(\theta, \theta') = \int P_1(\theta, \vartheta) P_2(\vartheta, \theta') d\vartheta.$$

- ▶ The MH error scales polynomially with the dimension; see blog post.

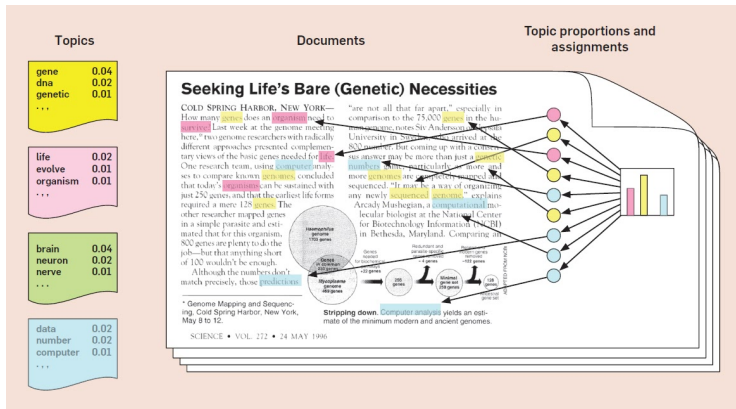
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- ▶ Consider MH with

$$q(\theta'|\theta) = \frac{1}{d} \sum_{k=1}^d \pi(\theta'_k|\theta_{\setminus k}) \mathbf{1}_{\theta'_k=\theta_k}, \quad \theta_{\setminus k} := (\theta_1, \dots, \theta_{k-1}, \theta_{k+1}, \dots, \theta_d).$$

- ▶ Then the probability of acceptance  $\alpha(\theta, \theta')$  is always 1.
  
- ▶ In practice, the systematic scan Gibbs sampler is more common, which consists in repeatedly: drawing  $\theta_1|\theta_{\setminus 1}$ , then  $\theta_2|\theta_{\setminus 2}$ , etc. always conditioning on the newest values available of each  $\theta_k$ .
- ▶ You can also partition  $\theta$  in arbitrary blocks.

# An example: Latent Dirichlet allocation



Fix a document index  $i \in [N]$ . By  $d$ -separation,

$$\pi_i | \pi_{\setminus i}, \mathbf{z}, \mathbf{y}, B = \pi_i | \mathbf{z}_i.$$

Now we recognize a conjugate model (see exercises), so that

$$\pi_i | \mathbf{z}_i \sim \text{Dir} \left( (\alpha_k + \sum_{\ell=1}^{L_i} \mathbf{1}_{z_{i\ell}=k})_{k \in [K]} \right).$$

## A Gibbs sampler for LDA: the conditional of $B$

By  $d$ -separation,

$$B|\pi, \mathbf{z}, \mathbf{y} \sim B|\mathbf{z}, \mathbf{y}.$$

By Bayes' theorem,

$$\begin{aligned} p(B|\mathbf{z}, \mathbf{y}) &\propto p(B, \mathbf{z}, \mathbf{y}) \\ &= p(\mathbf{y}|B, \mathbf{z})p(\mathbf{z}|B)p(B). \end{aligned}$$

Note that  $\mathbf{z} \perp B$  in the DAG, and

$$\begin{aligned} p(B|\mathbf{z}, \mathbf{y}) &\propto \prod_{i=1}^N \prod_{\ell=1}^{L_i} \prod_{k=1}^K B_{k\ell}^{1_{z_i\ell=k} 1_{y_i\ell=v}} \prod_{k=1}^K 1_{\Delta_v}(B_{k:}) \prod_{v=1}^V B_{kv}^{\gamma_v-1}. \\ &= \prod_{k=1}^K \left[ 1_{\Delta_v}(B_{k:}) \prod_{i=1}^N \prod_{\ell=1}^{L_i} B_{k\ell}^{1_{z_i\ell=k} 1_{y_i\ell=v}} \prod_{v=1}^V B_{kv}^{\gamma_v-1} \right]. \end{aligned}$$

The conjugacy should now be obvious, as well as the conditional independence of the rows of  $B$ . We see in particular that

$$B_{k:}|\mathbf{z}, \mathbf{y} \sim \text{Dir} \left( (\gamma_v + \sum_{i=1}^N \sum_{\ell=1}^{L_i} 1_{z_i\ell=k} 1_{y_i\ell=v})_{v \in [V]} \right).$$

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## A Gibbs sampler for LDA: the conditional of $z$

Fix  $i \in [N]$  and  $\ell \in [L_i]$ . Then by the DAG structure,

$$z_{i\ell} | \pi, \mathbf{z}_{\setminus i\ell}, \mathbf{y}, B \sim z_{i\ell} | \pi_i, y_{i\ell}, B.$$

Using Bayes' theorem,

$$\begin{aligned} p(z_{i\ell} = k | \pi_i, y_{i\ell}, B) &\propto p(z_{i\ell} = k, \pi_i, y_{i\ell}, B) \\ &= p(y_{i\ell} | z_{i\ell} = k, \pi_i, B) p(z_{i\ell} = k, B | \pi_i) p(\pi_i). \end{aligned}$$

Upon noting that  $y_{i\ell} \perp \pi_i | z_{i\ell} = k, B$  and  $z_{i\ell} = k \perp B | \pi_i$ , and keeping only terms that contain  $z_{i\ell}$ , it comes

$$p(z_{i\ell} = k | \pi_i, y_{i\ell}, B) \propto p(y_{i\ell} | z_{i\ell} = k, B) p(z_{i\ell} = k | \pi_i) \propto B_{ky_{i\ell}} \pi_{ik}.$$

To sample from the conditional of  $z_{i\ell}$ , it is thus enough to draw a new index  $k \in [K]$  with probability

$$\frac{B_{ky_{i\ell}} \pi_{ik}}{\sum_{k=1}^K B_{ky_{i\ell}} \pi_{ik}}.$$



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- ▶ Let  $S$  be a linear involution of  $\mathcal{X} \subset \mathbb{R}^{2d}$ , such that  $\eta \circ S = \eta$  for some (possibly unnormalized) PDF  $\eta$ .

- ▶ Let further  $\Phi : \mathbb{R}^{2d} \rightarrow \mathbb{R}^{2d}$  be a  $C^1$ -diffeomorphism such that  $S \circ \Phi = \Phi^{-1} \circ S$ .

- ▶ Now let

$$\alpha(x) \triangleq 1 \wedge \frac{\eta(\Phi(x))}{\eta(x)} |\Phi'(x)|, \quad (1)$$

and consider the Markov kernel

$$P_{aHMC}(x, A) = \alpha(x) 1_{\Phi(x) \in A} + (1 - \alpha(x)) 1_{S(x) \in A}.$$

### Proposition

$P_{aHMC}$  leaves  $\eta$  invariant.

## Hamilton's equations of motion

Consider a physical system described by Hamiltonian  $H(q, p)$  in phase space  $(q, p) \in \mathbb{R}^{2d}$ . Then the trajectories are prescribed by

$$\dot{q}_i = \frac{\partial H}{\partial p_i} \quad \dot{p}_i = -\frac{\partial H}{\partial q_i}. \quad (2)$$

- ▶ Given an initial point  $(q, p)$ , solve (2) and denote the corresponding position in  $\mathbb{R}^{2d}$  at time  $t > 0$  by  $\Phi_t(q, p)$ .
- ▶ (2) implies that  $t \mapsto H(\Phi_t(q, p))$  is constant.
- ▶ As an example, consider  $H(q, p) = \frac{1}{2}q^2 + \frac{1}{2}p^2$ .

- ▶ One idea would be to put some monotone function of the target in the Hamiltonian, such as  $H(q, p) = -\log \pi(q) + \frac{1}{2}p^T M p$ .
- ▶ We know approximations of the Hamiltonian flow, such as the leapfrog (aka velocity Verlet) integrator. It is defined as  $\psi_h^n = \psi_h \circ \dots \circ \psi_h$ , where  $(p', q') = \psi_h(p, q)$  is

$$\begin{aligned}p_{1/2} &= p + \frac{h}{2} \nabla \log \pi(q) \\q' &= q + h M^{-1} p_{1/2} \\p' &= p_{1/2} + \frac{h}{2} \nabla \log \pi(q');\end{aligned}$$

### Proposition

The leapfrog integrator satisfies  $S \circ \psi_h^n = (\psi_h^n)^{-1} \circ S$  for  $S(q, p) = (q, -p)$ , and  $|\det(\psi_h^n)'(q, p)| = 1$ .

## Hamiltonian Monte Carlo mimics a physical system

- ▶ Let  $\log \tilde{\pi}(q, p) = \log \pi(q) + \frac{1}{2}p^T M(q)p$ .
- ▶ Consider the Markov kernel  $\tilde{P}((q, p), (q', p'))$  given by the product of

$$p' \sim \mathcal{N}(0, M(q)^{-1})$$

and

$$P_{aHMC}(x, A) = \alpha(x)1_{\psi_h^n(x) \in A} + (1 - \alpha(x))1_{S(x) \in A}, \text{ where } x = (p, q).$$

where

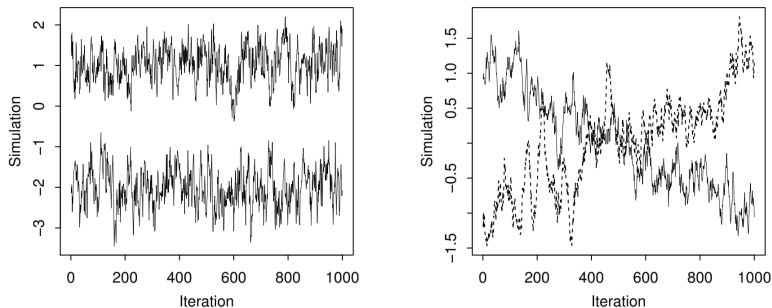
$$\alpha(x) \triangleq 1 \wedge \frac{\tilde{\pi}(\psi_h^n(x))}{\tilde{\pi}(x)} \frac{1}{|(\psi_h^n)'(x)|}, \quad (3)$$

Then  $\tilde{P}$  leaves  $\pi$  invariant.



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# What can go wrong?



**Figure:** Taken from (Gelman et al., 2013)

We need to monitor both cross-chain and within-chain behavior.

## Comparing $P$ chains with overdispersed starting points

- ▶ The behaviour of the  $P$  traces should become similar.
- ▶ Always make visual sanity checks!
- ▶ Scalar estimates should converge to the same value.
- ▶ We can also compare the variance of a scalar estimate within- and across chains

### The Gelman-Rubin diagnostic

- ▶ Choose an  $f$  of interest, e.g.  $f(\theta) = \theta_1$ .
- ▶ Compute  $B := \frac{T}{P-1} \sum_{p=1}^P (\bar{f}_{\cdot p} - \bar{f}_{\cdot\cdot})^2$ .
- ▶ Compute  $W := \frac{1}{P} \sum_{p=1}^P \left[ \frac{1}{T-1} \sum_{t=1}^T (\bar{f}_{tp} - \bar{f}_{\cdot p})^2 \right]$ .
- ▶ Then check whether

$$\hat{R} = \sqrt{\frac{\frac{T-1}{T} W + \frac{1}{T} B}{W}} \in [1, 1.1].$$

- ▶ See (Vats and Knudson, 2021) for an insightful discussion.

### Single-chain diagnostics

- ▶ The idea is to compare different chunks of a single chain.
- ▶ At stationarity, large chunks should be statistically indistinguishable.
- ▶ The **Geweke diagnostic** tests this similarity (Geweke, 1992)

### Effective sample size

- ▶ Autocorrelation in each chain is what increases the variance of scalar estimands, compared to i.i.d. draws from  $\pi$ .
- ▶ We can estimate this autocorrelation, and build an estimator for  $PT$  times the ratio of the two variances  $\widehat{ESS} \in [1, PT]$ , called, the **effective sample size**; see Section 11.5 of (Gelman et al., 2013).
- ▶ Vats and Knudson, 2021 note that

$$\hat{R} \approx \sqrt{1 + P/\widehat{ESS}},$$

so  $\hat{R} = 1.1$  **only** corresponds to  $\widehat{ESS} = 5P$ .

## Take-home message

- ▶ MCMC approximates the integrals in the expected utility framework.
  - ▶ Try to **leverage the problem's structure** to design your kernels.
  - ▶ Otherwise, try standard kernels like HMC.
  - ▶ Always monitor convergence.
- 
- ▶ HMC with NUTS is the default choice in most probabilistic programming frameworks.
  - ▶ MCMC is a **rich research topic**. Some keywords: Wang-Landau, Langevin, equi-energy, hit-and-run, bouncy particle sampler.
  - ▶ Besides Markov chains, checkout **sequential Monte Carlo samplers** (Del Moral, Doucet, and Jasra, 2006).
  - ▶ Deterministic methods are also investigated: **quasi-Monte Carlo methods** (Dick and Pillichshammer, 2010) have the best convergence rates as soon as the integrand is smooth.

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